

DONG-KYUM KIM

CONTACT INFORMATION

Max Planck Institute for Security and Privacy ([MPI-SP](#))

Website : kdkyum.github.io

Email : dong-kyum.kim@mpi-sp.org

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

2016 – 2022

Ph.D. in Physics

- Advisor : Prof. [Hawoong Jeong](#)
- Dissertation : Nonequilibrium Statistical Physics Study using Deep Learning

Seoul National University (SNU)

2011 – 2015

Bachelor of Science (BS) in Physics with a minor in Computer Science & Engineering

RESEARCH INTEREST

Artificial Intelligence (AI), Deep Learning, Machine Learning, Interpretable AI, Mechanistic Interpretability, Large Language Models (LLMs), Data Science, AI for Physics, Statistical Physics, Nonequilibrium Physics, Neuroscience, Brain-Inspired AI, Learning & Memory

EMPLOYMENT HISTORY

Max Planck Institute for Security and Privacy (MPI-SP)

Oct. 2024 – Present

Postdoctoral Researcher

- Hosted by prof. [Meeyoung Cha](#) (Director).
- Data Science for Humanity Group

Institute for Basic Science (IBS)

Mar. 2022 – Sep. 2024

Postdoctoral Researcher

- Hosted by prof. [Meeyoung Cha](#) (Chief Investigator).
- Data Science Group, Center for Mathematical and Computational Science

Samsung Electronics

Sep. 2017 – Dec. 2017

Data Science Intern

- Collaborated with [Daniel Kim](#) (Senior Data Scientist).
- Improved anomaly image classification tasks via distributed multi-GPU training methods of Keras & Spark.
- Implemented a distributed image searching framework to detect similar patterns in images through Elasticsearch.

PUBLICATIONS

[†] : equal contribution.

Minsung Kim, **Dong-Kyum Kim**, Jea Kwon, Nakyeong Yang, Kyomin Jung, and Meeyoung Cha. “How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models”. In : *arXiv preprint arXiv:2510.02370 (2025)*.

Dong-Kyum Kim, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. “Bilinear relational structure fixes reversal curse and enables consistent model editing”. In : *arXiv preprint arXiv:2509.21993 (2025)*.

Nakyeong Yang, **Dong-Kyum Kim**, Jea Kwon, Minsung Kim, Kyomin Jung, and Meeyoung Cha. “Erase or Hide ? Suppressing Spurious Unlearning Neurons for Robust Unlearning”. In : *arXiv preprint arXiv:2509.22263 (2025)*.

Yeongwoo Song, Jaeyong Bae, **Dong-Kyum Kim**, and Hawoong Jeong. “Uncovering Emergent Physics Representations Learned In-Context by Large Language Models”. In : *arXiv preprint arXiv:2508.12448 (2025)*.

Minsung Kim, Jea Kwon, **Dong-Kyum Kim**, and Meeyoung Cha. “In Search of the Engram in LLMs : A Neuroscience Perspective on the Memory Functions in AI Models”. In : *The Fourth Blogpost Track at ICLR 2025*.

Jea Kwon, Sunpil Kim, **Dong-Kyum Kim**, Jinhyeong Joo, SoHyung Kim, Meeyoung Cha, and C. Justin Lee. “SUBTLE : An unsupervised platform with temporal link embedding that maps animal behavior”. In : *International Journal of Computer Vision* (2024).

Gwangsu Kim, **Dong-Kyum Kim**, and Hawoong Jeong. “Spontaneous emergence of rudimentary music detectors in deep neural networks”. In : *Nature Communications* **15**, 148 (2024).

Dong-Kyum Kim[†], Jea Kwon[†], Meeyoung Cha, and C. Justin Lee. “Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity”. In : *Advances in Neural Information Processing Systems* (2023).

Sangyun Lee, **Dong-Kyum Kim**, Jong-Min Park, Won Kyu Kim, Hyunggyu Park, and Jae Sung Lee. “Multidimensional entropic bound : Estimator of entropy production for Langevin dynamics with an arbitrary time-dependent protocol”. In : *Physical Review Research* **5**, 013194 (2023).

Vyacheslav Shen, **Dong-Kyum Kim**, Elke Zeller, and Meeyoung Cha. “Neural Classification of Terrestrial Biomes”. In : *2023 IEEE International Conference on Big Data and Smart Computing (BigComp)*, pp. 163-166, (2023).

Youngkyoung Bae, **Dong-Kyum Kim**, and Hawoong Jeong. “Inferring dissipation maps from videos using convolutional neural networks”. In : *Physical Review Research* **4**, 033094 (2022).

Dong-Kyum Kim[†], Sangyun Lee[†], and Hawoong Jeong. “Estimating entropy production with odd-parity state variables via machine learning”. In : *Physical Review Research* **4**, 023051 (2022).

Dong-Kyum Kim and Hawoong Jeong. “Deep reinforcement learning for feedback control in a collective flashing ratchet”. In : *Physical Review Research* **3**, L022002 (2021).

Dong-Kyum Kim[†], Youngkyoung Bae[†], Sangyun Lee, and Hawoong Jeong. “Learning Entropy Production via Neural Networks”. In : *Physical Review Letters* **125**, 140604 (2020). [arXiv : 2003.04166 \[cond-mat.stat-mech\]](#).

Dong-Kyum Kim[†], Byunghwee Lee[†], Daniel Kim, and Hawoong Jeong. “Multi-label classification of historical documents by using hierarchical attention networks”. In : *Journal of the Korean Physical Society* **76**, 368 (2020).

AWARDS

- Pre-doctoral Fellow of Physics at KAIST

Aug. 30, 2021

ACADEMIC SERVICE

Reviewer

- [ICLR 2026 Main Conference](#)
- [NeurIPS 2025 Main Conference](#)
- [ICML 2025 Workshop on Assessing World Models](#)
- [ICLR 2025 Workshop on Representational Alignment \(Re-Align\)](#)
- [NeurIPS 2024 Workshop on Behavioral Machine Learning](#)
- [ICML 2024 Workshop on Large Language Models and Cognition](#)

PRESENTATIONS

Invited talks and lectures

- Computational Physics Course in KAIST (Daejeon, Korea). May. 1, 2023
“Deep learning applications : Nonequilibrium statistical physics study using AI”
- Computational Physics Course in KAIST (Daejeon, Korea). Apr. 24, 2023
“Deep Learning Introduction”
- IBS Winter School on AI-Boosted Basic Science (Daejeon, Korea). Dec. 13, 2022
“Resemblances between Transformer’s Nonlinearity and NMDA Receptor Dynamics”
- KIAS CAINS Summer Workshop (Jeju, Korea). Sep. 2, 2022
“Working and reference memory in transformers on a navigation task”
- KIAS Nonequilibrium Statistical Physics of Complex Systems (Seoul, Korea). Jul. 25, 2022
“Deep reinforcement learning for optimal mechanism in active Brownian particles”
- SNU Physics and AI Winter School (Seoul, Korea). Feb. 24, 2022
“Exploring Irreversibility via Machine Learning”

- APCTP Workshop for Physics and Machine Learning (Jeju, Korea). Nov. 26, 2021
“Exploring optimal mechanisms in active Brownian particles via deep reinforcement learning”
- Seoul National University Statistical Physics Seminar ((Online) Korea). Feb. 1, 2021
“Methods of estimating entropy production”
- Korean Physical Society Fall Meeting ((Online) Korea). Nov. 6, 2020
“Deep reinforcement learning for feedback-controlled flashing ratchets”
- NetSci2020 ((Online) Rome, Italy). Sep. 22, 2020
“Discovering wiring patterns of neural networks via backboning”
- Korean Physical Society Spring Meeting ((Online) Korea). Jul. 13, 2020
“Neural estimator for entropy production”
- Quantifying Success satellite at NetSci2019 (Burlington, Vermont, USA). May. 27, 2019
“Quantifying Individual Reputation in Large-scale Historical Documents”

IN THE PRESS

Selected list of media coverages

- “Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity” (*NeurIPS* 2023)
 - [IBS Research News \(2023/11/28\)](#); [Korean version](#).
 - [Donga Science \(2023/11/30\)](#).
 - [EurekAlert! \(2023/12/18\)](#).
- “Learning Entropy Production via Neural Networks” (*Phys. Rev. Lett.* **125**, 140604, 2020)
 - [Physics and High Technology \(2020/12/17\)](#).

REFERENCES

[Hawoong Jeong](#)

Professor
Department of Physics, KAIST
✉ hjeong@kaist.edu

[Meeyoung Cha](#)

Director
Data Science for Humanity Group, MPI-SP
✉ mia.cha@mpi-sp.org

[Yongjoo Baek](#)

Professor
Department of Physics & Astronomy, SNU
✉ y.baek@snu.ac.kr

[C. Justin Lee](#)

Director
Center for Cognition and Sociality, IBS
✉ cjl@ibs.re.kr

[Junghyo Jo](#)

Professor
Department of Physics Education, SNU
✉ jojunghyo@snu.ac.kr